

Theorem 1. *Deadlock-free routing can be constructed for a network $G = (V, E)$ if and only if G contains two edge-disjoint directed trees rooted at the same vertex v , one tree directed into v and the other directed away from v .*

Sufficiency of the two-tree condition of **Theorem 1** for the existence of deadlock-free routing follows from the fact that given such trees, any message can be routed by first sending it to the common root v along the first tree, and then sending it to its target along the second tree.

The two-tree condition was known to be sufficient for deadlock-free routing. For example, [BS92; JMS94] established this fact, while [Aba97, Section 4.4] uses such two trees to construct a deadlock-free routing for a 4x4 mesh. However, to the best of our knowledge, the necessity of this condition has neither been recognized nor proven.

To prove **necessity** we explicitly construct the two required trees from a network with a given deadlock-free routing. Using the result of [DS87], the dependency graph induced by the deadlock-free routing is acyclic, so there is a dependency-respecting total ordering of the graph edges E such that every vertex is reachable from every other vertex through a sequence of edges that are ascending according to the total order.

Given a total ordering of the edges, we introduce three definitions: a global attractor, the attraction number and the attraction subgraph.

Definition. *A vertex in a directed graph is called a **global attractor** if it is reachable from all other vertices.*

Definition. *Given a strongly connected directed graph $G = (V, E)$ with a total ordering of its edges $E = (e_1, \dots, e_m)$, we define its **attraction number** s to be the smallest integer such that there exists a vertex $v \in V$ reachable from all other vertices using only the edges in the prefix $\{e_1, \dots, e_s\}$.*

The existence of such an s follows from the fact that G is strongly connected. We note that the attraction number is the minimal size of the prefix of the edges needed for the graph to have a global attractor. For the purpose of this definition, reachability using edges $\{e_1, \dots, e_s\}$ is not restricted to paths that respect the total ordering of the edges.

Definition. *Given a strongly connected directed graph $G = (V, E)$ with a total ordering of its edges $E = (e_1, \dots, e_m)$, we define its **attraction subgraph** to be the subgraph $G_s = (V, E_s)$ with the same vertices but only the first s edges in E : $E_s = (e_1, \dots, e_s)$, where s is the attraction number of the graph.*

By definition, the attraction subgraph has a global attractor.

Lemma 1.1. *Let $G = (V, E)$ be a strongly connected directed graph with a total ordering of its edges $E = (e_1, \dots, e_m)$ that is consistent with the dependencies induced by a deadlock-free routing.*

*If there is only a single global attractor v in the attraction subgraph G_s , then the two-trees required by **Theorem 1** exist in G .*

Proof. If v is the only vertex reachable from all vertices in G_s , no other vertex in G_s can be reachable from it, so there are no outbound edges of v in G_s . Therefore, routing